

Application of Advanced Statistical Learning Models for a Data Analysis Problem (40% in total and will be adjusted based on peer review)

This is a group assignment. You will gain more hands-on experience and a deeper understanding of the various predictive and deep neural network models, through working on real-world data and data analytics problem. **Models not covered in the module can be used as well with the right justification.**

For this assignment, all groups will be provided a multi-modal clinical dataset extracted from the **MIMIC-IV (Medical Information Mart for Intensive Care) database V3.1**, which was just newly released in 2023. The extracted dataset will contain various time-series and text-based clinical and physiological data of 94,458 patients. You are required to build a **multi-modal predictive model** to estimate the patients' time-to-discharge from the **Intensive Care Unit (ICU)**. The objective of this assignment is to evaluate whether the students can:

- 1) Develop and implement model(s) extraction from multi-modal data, including static, time-series and text data.
- 2) Develop and implement model(s) to predict a time-to-event outcome.
- 3) Analyze and explain the differences in performance of different models.

Note: NUS takes plagiarism very seriously. A student should not copy from seniors' assignments of previous years nor his/her own assignments of other courses.

Group and individual efforts will be assessed in this assignment. All groups are to perform the following tasks for the assignment:

Group Assessment Component (40% of marks)

- 1) Pick appropriate models for the right type of data, and optimize the time-to-event model developed to achieve reasonable predictive accuracy. (10% of the total marks)

Deadline: 17th April 2026

Award up to 10% when groups:

- Apply any pre-processing needed to get the data ready. (2.5%)
- Apply and optimise appropriate models to capture information from various forms of clinical data. (5%)
- Integrate information from the above models. (2.5%)
- Details of the above items should be included in written report in the following component.

(Note: All codes of the implemented models must be uploaded to GitHub and the GitHub URL must be included in the report. This total 10% of marks will be deducted if the group fails to submit the codes before the deadline.)

- 2) In a written report, summarise the finalised predictive model and all its building components: clearly explain all the deployed models, the pre- and post-data processing steps and illustrate the rationale and inspiration behind the finalised model. The report must be uploaded onto the Canvas Submission Folder. The report is recommended to be no more than 10 pages. (10% of the total marks)

Deadline: 17th April 2026

Award up to 10% when groups:

- Clearly present and explain the models used in the assignment. (2.5%)
- Clearly explain the rationale behind the choice of models and optimisation methods. (2.5%)
- Clearly illustrates the results with appropriate use of graphics. (3%)
- Write in a clear and concise manner. (2%).

- 3) Individual groups will present their final models and findings at an oral presentation. The groups will be assessed based on the understanding of the problem and the developed model.

Deadline: 17th April 2026

Award up to 20% when student:

- Clearly presents and explains their work.
- Answers questions related to their problem statement and selected model.

The detailed grading rubrics are as follows:

Content/ Criteria (equal weightage)	Band 1 criteria	Band 2 criteria	Band 3 criteria
Clarity (5%)	Audience can follow but with effort. A few of the key concepts are clearly presented.	Most of the presentation is clear but a few points are confusing. Most of the key concepts are clearly presented.	Audience can follow easily. Key concepts are clearly presented.
Critical Thinking (5%)	Demonstrates some critical thinking in the evaluation of findings and recommendations.	Demonstrates good critical thinking in the evaluation of findings and recommendations.	Demonstrates strong critical thinking in the evaluation of findings and recommendations.
Quality of Presentation Slides (5%)	Slides are somewhat cluttered but readable.	Slides are uncluttered, readable and somewhat innovative.	Slides are uncluttered, easy to read and strongly innovative.
Presentation Style (3%)	Much of the information is read.	Some reference to slides as aide de memoire.	Presentation is paced for audience.

Timeliness (2%)	Overruns	Rushes some parts to finish on time.	Completes on time.
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Peer Review Adjustment

Deadline: 17th April 2026

Final marks of individual member will be adjusted based on the peer review. The standard adjustment framework developed by the school will be employed.